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Roll No .....

**FT-404 (GS)**  
**B.E. IV Semester** Examination, June 2020  
**Grading System (GS)**  
**Electrical Fire Safety**  
**Time : Three Hours**

**Maximum Marks : 70**

**Note:** i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Give a typical layout of a power system between generation and use of electrical power.  
b) Explain clearly the difference between feeder, distributor and service mains?
2. Calculate the kVA taken by a 10 km long, 3-phase, 3-core cable, if the capacitance measured between any two cores is  $0.3 \mu\text{F/km}$  when it is connected to 10 kV, 50 Hz bus-bar?
3. a) How will you select location for a Large HV sub-station?  
b) Give method for fighting a fire on large oil cooled transformer.
4. Enlist General safety precautions taken care in handling with electric systems.
5. What is meant by Corona? What are the various factors which affect Corona? How it can be minimised? Discuss the advantages and disadvantages of Corona.
6. Why earthing is important? What are the different types of earthing? Explain. Mention any one with neat diagram.

OR

Why cooling is necessary in electrical transformer? Explain different types of cooling methods used in transformer.

7. Explain in detail about IS-2148 for flame proof enclosures of electrical apparatus.

OR

Explain Buchholz relay. Why it is used in transformer?

8. Answer any two of the following:

- a) What are the General Nuclear Safety objectives? Explain.
- b) Enlist General difference between overhead transmission system and under-ground cable system used for electric power transmission.
- c) Describe safety instructions used with operating high voltage apparatus.
- d) What are the difference between a fuse wire and circuit breaker?

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